

GNU nano 7.2

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#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    printf("Parent process started. PID = %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("Child process running. PID = %d\n", getpid());

        sleep(2);

        printf("Child process completed.\n");

        exit(5);
    }
    else
    {
        // Parent process
        printf("Parent is waiting for child...\n");

        waitpid(pid, &status, 0);
    }
}
```

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        printf("Child process completed.\n");
        exit(5);
    }
    else
    {
        // Parent process
        printf("Parent is waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status))
        {
            printf("Child exited with status %d\n",
                   WEXITSTATUS(status));
        }

        printf("Parent process completed.\n");
    }

    return 0;

```

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sravya@Sravya1955:~/os$ nano waitpid_demo.c
sravya@Sravya1955:~/os$ nano waitpid_demo.c
sravya@Sravya1955:~/os$ gcc waitpid_demo.c -o waitpid_demo
sravya@Sravya1955:~/os$ ./waitpid_demo
Parent process started. PID = 2391
Parent is waiting for child...
Child process running. PID = 2393
Child process completed.
Child exited with status 5
Parent process completed.
sravya@Sravya1955:~/os$ gcc waitpid_demo.c -o waitpid_demo
sravya@Sravya1955:~/os$ ./waitpid_demo
Parent process started. PID = 2412
Parent is waiting for child...
Child process running. PID = 2413
Child process completed.
Child exited with status 5
Parent process completed.
sravya@Sravya1955:~/os$ |

```

```

Child exited with status 5
Parent process completed.
sravya@Sravya1955:~/os$ ls
1          copy_low.txt      filecopy.c      memory.c        os1.1.7.c      os6.c          quotes        signal2.c      std.c.save
1.1.1c    copy_standard.txt    filecopy1       mmap.txt        os1.1.8.c      os7.c          quotes.c      signal3        symlink.txt
1.c       cow                  filecopy1.c     mmap_file       os1.1.c        os8.c          quotes.sh     signal3.c     test.txt
2         cow.c              filecopy2       mmap_file.c     os1.2.c        output.txt     redirect      signal4        test1.txt
2.c       destination.txt      filecopy2.c     new            os1.3.c        parser         redirect.c    signal4.c     test2.txt
3         destination2.txt     filecopy_std    op.save        os1.4.c        parser.c       sample.txt    signal5        tokenizer
3.c       destination_std.txt  filecopy_std.c  os.c           os1.c          parser1        shell         signal5.c     tokenizer.c
4.c       docs               hardlink.txt    os1.1.1.c      os1.xc         parser1.c      shell.c       signal6        waitpid_demo
'Another File.txt' escape          input.txt       os1.1.2.c      os14.c         process        shell_history signal6.c     waitpid_demo.c
FileDescriptors escape.c        memory          os1.1.3.c      os15.c         process.c      shell_history.c signal7        src
LinuxLab  exec             memory.c        os1.1.4.c.save os2.c          program1.c    shell_history.c.save signal7.c
'My File.txt' fifo_project    memory1.c       os1.1.5.c      os3.c          program7        signal1       std
MyProject file1.txt       memory1.c       os1.1.6        os4.c          program7.c     signal1.c     src
bin       filecopy         memory          os1.1.6.c      os5.c          project        signal2       std.c
sravya@Sravya1955:~/os$ echo $PATH
/home/sravya/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files
/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/WIND
OWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/c/Program Fil
es/dotnet:/mnt/c/Program Files/PuTTY:/mnt/c/Users/chand/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/chand/AppData/Local/Microsoft/WindowsApps/mn
t/c/Users/chand/Downloads/38833FF26BA1D.UnigramPreview_g9c9v27vpyspw!App/gcc_15.1.0_gdb_16.3.90_20250511_binutils_2.44_mingw_w64_v12.0.0/bin:/mnt/d/Users/ch
and/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/Java/jdk-21/bin:/mnt/c/Users/chand/AppData/Local/Programs/Microsoft VS Code/bin:/snap
bin
sravya@Sravya1955:~/os$ |

```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

int main(int argc, char *argv[])
{
    char *path;
    char *path_copy;
    char *dir;
    char full_path[PATH_MAX];

    if (argc != 2)
    {
        printf("Usage: %s <command>\n", argv[0]);
        return 1;
    }

    path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    path_copy = strdup(path);

    if (path_copy == NULL)
    {
        perror("strdup");
        return 1;
    }

    dir = strtok(path_copy, ":");
```

```

while (dir != NULL)
{
    snprintf(full_path, sizeof(full_path), "%s/%s",
             dir, argv[1]);

    if (access(full_path, X_OK) == 0)
    {
        printf("Command found: %s\n", full_path);
        free(path_copy);
        return 0;
    }

    dir = strtok(NULL, ":");
}

printf("Command '%s' not found in PATH.\n", argv[1]);

free(path_copy);

return 1;
}

```

```

bin
sravya@Sravya1955:~/os$ nano path_resolver.c
sravya@Sravya1955:~/os$ gcc path_resolver.c -o path_resolver
sravya@Sravya1955:~/os$ ./path_resolver ls
Command found: /usr/bin/ls
sravya@Sravya1955:~/os$ ./path_resolver hello123
Command 'hello123' not found in PATH.
sravya@Sravya1955:~/os$ gcc path_resolver.c -o path_resolver
sravya@Sravya1955:~/os$ ./path_resolver ls
Command found: /usr/bin/ls
sravya@Sravya1955:~/os$ ./path_resolver hello123
Command 'hello123' not found in PATH.
sravya@Sravya1955:~/os$ |

```